① ⓐ Nominal, ⓑ Ordinal, ⓒ Continuous.

② ⓐ $E(\hat{p}_i) = \frac{n p_i}{n} = p_i$, ⓑ $Var(\hat{p}_i) = \frac{n p_i (1-p_i)}{n^2} = \frac{p_i(1-p_i)}{n}$

ⓑ $P(X_1 = x_1, X_2 = x_2) = \frac{\lfloor n}{\lfloor x_1 \lfloor x_2 \lfloor n-x_1-x_2} p_1^{x_1} p_2^{x_2} (1-p_1-p_2)^{n-x_1-x_2}$, $x_1 \geq 0$

$x_2 \geq 0$

$x_1 + x_2 \leq n$

ⓒ $E(X_1X_2) = \sum\limits_{\substack{x_1, x_2 \geq 0 \\ x_1+x_2 \leq n}} \sum \frac{x_1 x_2 \lfloor n}{\lfloor x_1 \lfloor x_2 \lfloor n-x_1-x_2} p_1^{x_1} p_2^{x_2} (1-p_1-p_2)^{n-x_1-x_2}$

$= \sum\limits_{\substack{x_1-1, x_2-1 \geq 0 \\ x_1-1+x_2-1 \leq n-2}} \sum \frac{p_1 p_2 n(n-1) \lfloor n-2}{\lfloor x_1-1 \lfloor x_2-1 \lfloor (n-2)-(x_1-1)-(x_2-1)} p_1^{x_1-1} p_2^{x_2-1} (1-p_1-p_2)^{(n-2)-(x_1-1)-(x_2-1)}$

$= n(n-1) p_1 p_2$

$cov(X_1, X_2) = E(X_1X_2) - E(X_1)E(X_2) = -n p_1 p_2$

$cov(\hat{p}_1, \hat{p}_2) = \frac{-p_1 p_2}{n}$

ⓓ $Var(\hat{p}_1 - \hat{p}_2) = Var(\hat{p}_1) - 2cov(\hat{p}_1, \hat{p}_2) + Var(\hat{p}_2)$.

$= \frac{(p_1+p_2) - (p_1-p_2)^2}{n} = \sigma^2$

ⓔ $Z = \sqrt{n}\sqrt{\frac{3}{2}}(\hat{p}_1 - \hat{p}_2)$ under H_0, $p_1 = p_2$

$\sigma^2 = \frac{2}{3n}$

Reject H_0 for $|Z| \geq z_{1-\alpha/2}$